

Jayanth Vallabi

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SUMMARY

Python Developer with **4+ years** of experience designing and deploying **production-grade backend systems, REST APIs, and AI/ML services** across **Healthcare, Finance, and Technology** domains. Expert in **Python, FastAPI, SQL, PySpark, Docker, Kubernetes, and AWS**, with strong skills in **microservices architecture, REST APIs, GraphQL, gRPC, and CI/CD automation**. Proven impact: **87% faster execution, 39% lower API latency, 99.5% uptime** across high-scale systems serving **1M+ records**. Additional expertise in **GenAI, LLMs, and MLOps** for AI-powered applications, with experience mentoring engineers and leading Agile cross-functional delivery.

SKILLS

Programming Languages: Python, SQL, PySpark, Bash, JavaScript

Python Libraries: FastAPI, Flask, Django, Pydantic, SQLAlchemy, Celery, asyncio, NumPy, Pandas, scikit-learn, PyTorch, TensorFlow, XGBoost, pytest

Backend Development: REST APIs, GraphQL, gRPC, Microservices Architecture, API Design, Authentication (OAuth, JWT), Asynchronous Programming, Object-Oriented Programming (OOP), Design Patterns

Databases: PostgreSQL, MySQL, MongoDB, Redis, Snowflake, ChromaDB, Pinecone, FAISS, Weaviate

Cloud & DevOps: AWS (SageMaker, S3, Lambda, EC2), Docker, Kubernetes, Apache Airflow, CI/CD, GitHub Actions, Git, Linux

AI/ML & MLOps: MLOps, MLflow, Model Versioning, Experiment Tracking, Model Monitoring, Model Deployment, Drift Detection, Feature Engineering, Hyperparameter Tuning, A/B Testing, Batch Inference, Real-time Inference

Generative AI & LLMs: Generative AI (GenAI), Large Language Models (LLMs), LLM Fine-tuning, LoRA, PEFT, Prompt Engineering, Hugging Face Transformers, LangChain, LlamaIndex, Retrieval-Augmented Generation (RAG), RAGAS, OpenAI API, Embeddings

Software Engineering Practices: Agile, Scrum, Jira, Test-Driven Development (TDD), Code Reviews, Technical Design, Documentation, Cross-functional Collaboration, Mentoring, HIPAA Compliance

EXPERIENCE

Software Development Engineer, AI/ML

Cigna Healthcare Inc.

October 2024 - Present, USA

- Designed and developed **Python-based AI/ML services** and **REST APIs** using **Python, FastAPI, scikit-learn, PySpark, and SQL**, powering **healthcare analytics, predictive modeling, and patient risk classification** across **1M+ member records**.
- Engineered and deployed scalable **real-time AI/ML inference APIs** and **microservices** using **FastAPI, Docker, Kubernetes, Apache Airflow, and AWS SageMaker**, reducing manual model execution from **4 hours to under 30 minutes** per release cycle.
- Implemented end-to-end **MLOps and CI/CD workflows** using **MLflow, GitHub Actions, Snowflake, Docker, and Kubernetes**, improving model release frequency from **monthly to bi-weekly** with automated **monitoring and drift detection**.
- Built **GenAI and LLM-powered healthcare document intelligence systems** using **Hugging Face Transformers, LangChain, RAG, prompt engineering, LoRA/PEFT, and vector databases**, cutting clinical document review time by **6 to 8 hours** per week while ensuring **HIPAA-compliant PHI handling**.
- Optimized **model serving and inference performance** using **Python profiling, batch inference, asynchronous programming, caching, and feature engineering**, lowering average inference latency from **900ms to 550ms**, while mentoring **3 junior AI/ML engineers**.
- Partnered with **data scientists, engineers, and healthcare stakeholders** across **Agile teams**, leading **technical design reviews** and **AI/ML model governance** to maintain **99.5% service availability** in cloud environments.

Software Engineer

Capgemini

May 2021 - December 2023, India

- Developed **Python-based backend services** and **REST APIs** for financial services clients using **Python/Golang, FastAPI, SQL, PySpark, scikit-learn, PyTorch, TensorFlow, and cloud platforms**, supporting **risk scoring, transaction analytics, and customer intelligence applications**.
- Engineered production-grade **backend modules** for **classification, forecasting, NLP, and anomaly detection** using **Python, feature engineering, time series analysis, ensemble learning, and deep learning**, improving system accuracy by **10 to 15%** over baseline.
- Automated **backend data and deployment pipelines** using **Python, Docker, Kubernetes, Apache Airflow, MLflow, Snowflake, and CI/CD**, reducing deployment effort by **12 to 15 engineering hours** per sprint.
- Built **Python-powered financial document search applications** using **FastAPI, LangChain, LlamaIndex, Hugging Face, ChromaDB, Pinecone, FAISS, and Weaviate**, improving API retrieval response time to **under 3 seconds**.
- Designed and integrated **backend services** and **microservices architecture** using **FastAPI, REST APIs, GraphQL, and gRPC**, enabling cross-system access for **5+ downstream enterprise applications**.
- Validated **backend outputs** using **pytest, experiment tracking, RAGAS, A/B testing, and stakeholder review**, reducing manual QA effort by **20+ hours** per month while supporting **finance compliance requirements**.

PROJECT

Healthcare Document Intelligence Platform

September 2025 - February 2026

- Built a **Python-based document intelligence platform** using **FastAPI, LangChain, Hugging Face Transformers, RAG, and Pinecone vector database**, enabling semantic search and Q&A over **10,000+ clinical documents** with retrieval response time under **2 seconds**.
- Developed **REST APIs and microservices** using **FastAPI, Pydantic, SQLAlchemy, and PostgreSQL** with **JWT authentication, asynchronous programming, and OOP design patterns**, supporting **500+ concurrent API requests** with automated **pytest** test coverage.
- Deployed the application as a **containerized microservice** using **Docker, Kubernetes, AWS Lambda, and CI/CD with GitHub Actions**, achieving **99% uptime** and zero-downtime deployments with automated monitoring.

EDUCATION

Master of Science in Computer Science

California State University Fullerton • USA • 2026 • 3.6